

1. Use the *syms* or *syms* command to create the following symbolic expressions:

- (a) $ex1 = x^2 - 1$
- (b) $ex2 = a * x^3 + b * x^2 + c * x + d$
- (c) $ex3 = \sin(x)$
- (d) $eq1 = x^2 = 1$
- (e) $eq2 = a * x^2 + b * x + c = 0$
- (f) $eq3 = (X + 1)^2 = 0$
- (g) $a = 6/5 + 4/3$, using both numerical and symbolic computation.

2. Use the variables defined in Practice Exercises 1 in these exercises.

- (a) Multiply $ex1$ by $ex2$, and name the result $y1$.
- (b) Divide $ex1$ by $ex2$, and name the result $y2$.
- (c) Use the *numden* function to extract the numerator and denominator from $y1$ and $y2$.
- (d) Use the *factor*, *expand*, *collect*, and *simplify* functions on $y1$ and $y2$.

3. Define a symbolic equation representing each equation in the given system of equations. Use the *solve* function to solve for x , y , and z of the following equation:

$$\begin{aligned}5 * x + 6 * y - 3 * z &= 10 \\3 * x - 3 * y + 2 * z &= 14 \\28x - 4 * y - 12 * z &= 24\end{aligned}$$

4. Use the variables defined in Practice Exercises 1 in these exercises. Use the *solve* command to solve the $eq1$, $eq2$ and $eq3$.

5. Using the *subs* function, substitute 4 into each expression/equation defined in Practice Exercises 1 for x .

6. Create a symbolic function called g for the following:

$$x^2 + \sin(x) * x$$

and use it to evaluate $g('a')$, $g(3)$, and $g([1 : 5])$.

7. Plot the functions:

- (a) Use *ezplot* to plot $\cos(x)$ from -2π to $+2\pi$.
- (b) Use *ezplot* to create an implicit plot of $x^2 - y^4 = 5$.
- (c) Use *ezplot* to create a parametric plot of $x = \sin(t)$ and $y = 3 \cos(t)$.

8. Find the first derivative with respect to x of the following expressions:

- (a) $x^2 + x + 1$
- (b) $\sin(x)$
- (c) $\ln(x)$
- (d) $\tan(x)$

9. Find the first partial derivative with respect to x of the following expressions:

- (a) $ax^2 + bx + c$
- (b) $\sin(xy)$
- (c) $x^{\frac{1}{2}} - 3y$
- (d) $\tan(x + y)$

10. Integrate the following expressions with respect to x :

- (a) $3x^3 - 5x^2 + 3$

- (b) $4x^3 + \cos(2x) + 3$
- (c) $4e^{-xt} + \sin(3t)$
- (d) $\sin(3x) + x \cos(x)$

11. Integrate the following expressions with respect to x :

- (a) $3x^3 - 5x^2 + 3$; limits are $(3, 5)$
- (b) $9xe^x$; limits are $(0, 5)$
- (c) $4e^{-xt} + \sin(3t)$; limits are $(0, \pi)$
- (d) $\sin(3t) + x \cos(x)$; limits are $(0, \pi)$